WASP-50b

R-1008

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-50_161212` · created 2026-07-30T02:30:02+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2457734.678 +/- 0.0025 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1749 +/- 0.0062
Transit depth (Rp/Rs)^2	3.06 +/- 0.22 [%]
Orbital Inclination (inc)	89.72 +/- 0.62
Airmass coefficient 1 (a1)	1.189 +/- 0.032
Airmass coefficient 2 (a2)	-0.1037 +/- 0.0094
Scatter in the residuals of the lightcurve fit is	2.66 %
Best Comparison Star	None
Optimal Aperture	3.97
Optimal Annulus	15.84
Transit Duration (day)	0.09689 +/- 0.00065

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1749 ± 0.0062 vs 0.1404 (deviation 3.04σ)
Duration fit vs published	0.09689 d vs 0.0754 d (deviation 18.05σ)
Mid-time O–C	67.6 min
Depth SNR	15.4
Residual scatter	2.66 %

Light curve

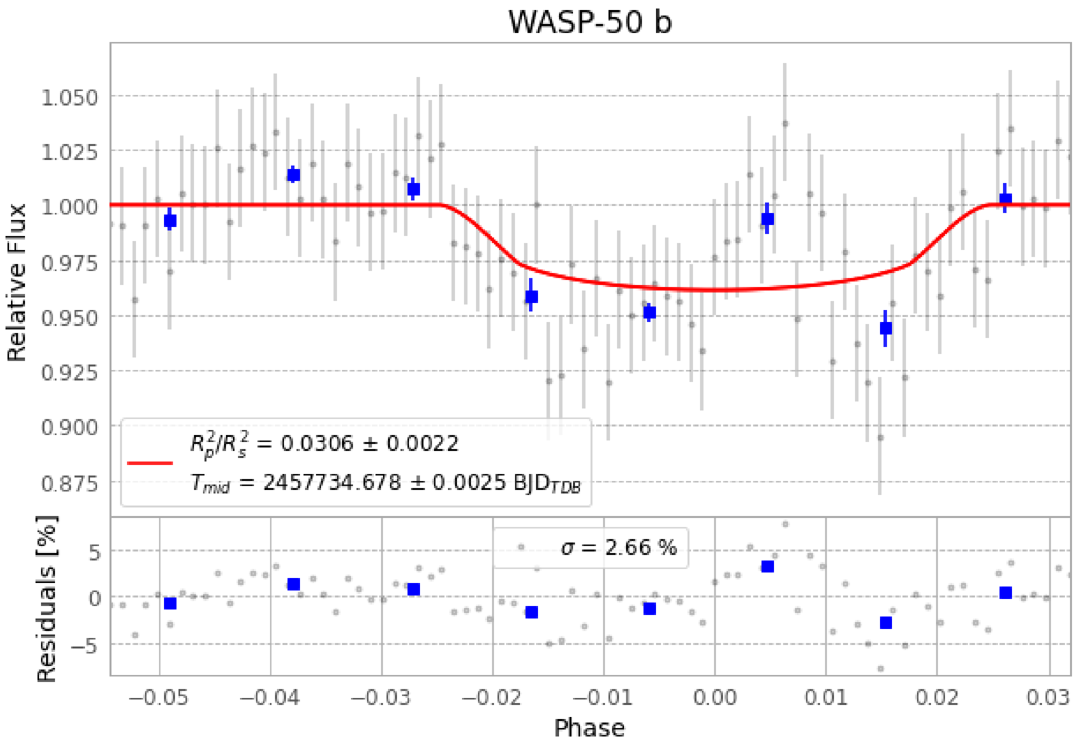

Figure 1 — FinalLightCurve_WASP-50 b_2016-12-11.png (SHA-256 33092add567b3040…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [297, 91], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 a5985904810f855e…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9`

Data provenance

82 FITS frames (54 MB), `WASP-50161212013612.FITS` → `WASP-50161212053912.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-50-161212.log (6e0699710fff…), FinalLightCurve_WASP-50 b_2016-12-11.png (33092add567b…), FinalLightCurve_WASP-50 b_2016-12-11.csv (445a0853b56d…)

Compute resources

Wall time 130.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)